

Kryptos K4 Solution and Procedure

To Yahoo User Group from Ibn Şifre,

A Forward: The Shadxws:

The world of spies is a world of shadows and mirrors. A world in which truth is elusive. A world where much of the data gathered is an incomplete picture of the total situation, and riddled with mistakes. This elusiveness, I believe, is one of the main themes of Kryptos. Below, I report what I have found. The deciphering is verifiable, and the process is repeatable. But, unfortunately in this world there is no decipherment algorithm that tells whether or not what is reported as plaintext is the truth - or a lie - or a bit of both. The decipherment and interpretations of what is found are mine and mine alone. I might, inadvertently, imply that a villain is good or a hero is a villain. I will do my best, but undoubtedly what you will find below is riddled with mistakes. As the Creator is the Most Merciful, the Most Forgiving, I must beg your pardon.

One other note: please keep in mind that decipherment of Kryptos reads like a spy novel, filled with riddles, sudden realizations, plot twists, multiple interpretations of information, different parties claiming that they are telling the truth and the other is lying, and a significant difficulty of even knowing who is who.

In this paper the following notation is used for indicating how a text is encrypted or decrypted. The cipher is a Vigenere Cipher which uses the KRYPTOS alphabet as was used in K1 and K2:

For encryption: $ENC(key, plaintext) = ciphertext$
For decryption: $DEC(key, ciphertext) = plaintext$

Examples:

$key = ELYOIE, plaintext = BERLIN, ciphertext = NYPVTT$
 $ENC(ELYOIE, BERLIN) = NYPVTT$
 $DEC(ELYOIE, NYPVTT) = BERLIN$

For ease one could use one of the interactive keyed Vigenère solvers on the web like Rumkin's Keyed Vigenère Cipher at <http://rumkin.com/tools/cipher/vigenere-keyed.php>

Now another point about shadows: That the steganographic clues of Kryptos' ciphertext seem to indicate the following:

Encrypting 'ENDY' of ENDYAHR with key 'V' yields:

$ENC(V, ENDY) = AISX \rightarrow A IS X$

And since 'ENDYA' is quite close to 'ANDYES' of John 8:32, 'And Ye Shall Know the ...' take 'DYES' and also decrypt with key V :

DEC(V, DYES) = HSID -> D IS H (backwards)

But in other places, we have many reasons to think that D opposes H. Also, the Stenographic clues of Kryptos imply that S and W go together, which give us:

S and W
H ~ D
A is X

which is an anagram of SHADXW, or SHADOW with the X covering the O. The Cyrillic Projector, Kryptos' sister work, uses the word TEHb, or shadow in Russian, as its cryptographic key; the pronunciation of which in English sounds very close to the word "ten". The number ten being X in Roman numerals suggest the following interpretation:

A is: X -> TEN -> TEHb -> Shadow

A is Shadow.

The Palimpsest as a One Time Pad:

Start with K3 and arrange it in 7 rows by 48 columns as shown in Fig 1-1.

	1	2	3	4
	123456789012345678901234567890123456789012345678			
1	ENDYAHROHN	LSRHEOCPT	EOIBIDYSH	NAIACHTNREYULDSL
2	NOHSNOSMR	WXMN	ETPRNGAT	IHNRRARPESLNNELEBLPIIACAEWMT
3	WNDITEEN	NRAHCTE	NEUDRETN	HAEOETFOLSEDTIWENHAEIOYTEY
4	QHEENCTAY	CREIFTB	RSPAMHHE	WENATAMATEGYEERLBTEEFOAS
5	FLOTUETUA	EOTOARMA	EERTNRTI	BSEDDNIAAHTTMSTEWPIEROA
6	GRIEWFEB	AECTDDH	ILCEIHSI	TEGEOAOSDDRYDLORITRKLML
7	AGTDHARD	PNEOHMG	FMFEUHEE	CDMRIPFEIMEHNLSSTTRTVDOHW

Fig. 1-1, K3

Flip K3 about the columns so that column 48 is first and column 1 is last, as shown in Fig 1-2. This is done only to make the techniques easier to visualize. Call it K3'.

	4	3	2	1
	876543210987654321098765432109876543210987654321			
1	LLSL	LDLU	YERN	THCA
2	TMWE	ACAI	IPLB	ELENNL
3	YETYO	IEAH	NEWIT	DESLOF
4	SAOF	EETBL	REEYGE	TAMATANE
5	AORE	IPWETS	MTTHAA	INDDES
6	HELM	LKRTI	ROLDYR	DDSOAEO
7	WHOD	VTRTT	SSLNHE	MIEFP

Fig. 1-2, K3'

Using the END - YAR (RAY) clue, place K3' in the form of a transverse EM wave with a period of 16 characters and an amplitude of 11 characters, as shown in Fig 1-3a.

	4	3	2	1
	876543210987654321098765432109876543210987654321			
4	LLSD	RNTH	NHSY	OETP
5	EACA	BELE	EPRA	TAGN
6	YOIE	WITD	FTEO	TERD
7	FEET	EYGE	TANE	MAPS
1	LLSE	IPWL	UYET	THACAI
2	TMWM	LKRI	IPLD	YRNNL
3	YETD	VTRA	HNEL	NHEESL
4	SAO	BLRE	TAMA	WEHH
5	AOR	ETSM	AIND	ITRN
6	HEL	TIRO	DDSO	ETIS
7	WHO	TTSS	MIEF	DCEE

Fig. 1-3a,

Collapse the wave so that all the columns are filled as shown in Fig 1-3b.

	1	2	3	4
	123456789012345678901234567890123456789012345678			
1	LLSEIPWL	UYETTHACAI	ADESBD	IBITREECOEHOTOEHORHUTOIE
2	TMWMLKRII	PLLDYRNNLSAEOGRN	HIHI	ECRPTEDTCERMSOWEIRN
3	YETDVTRAHNELN	HSEESLOPIRMEAHNHUEFUENEHOENRNEEHDTGW		
4	SAOLLSDBLRERN	THHTAMANHSYWEH	HOETPRBTFRSLNYATCAYDNQ	
5	AOREACAETSM	BELEA	INDEPRAITRNTAGNAMRANMXWAUTENSHOF	
6	HELIOIETIROWIT	DDDSOFTEOET	ISTERDLIHDTCHAABEFTIDNG	
7	WHOFEETTT	SSEYGEMIEFTANED	CEEMAPSMFGMIERCPDRANEEHA	

Fig. 1-3b

Use only rows 25 through 48 of Fig 1-3b as 'W', shown in Fig 1-4. This forms our palimpsest.

	1	2
	123456789012345678901234	
1	LLSEIPWL	UYETTHACAI
2	TMWMLKRII	PLLDYRNNLSAEOGR
3	YETDVTRAHNELN	HSEESLOPIRME
4	SAOLLSDBLRERN	THHTAMANHSYW
5	AOREACAETSM	BELEA
6	HELIOIETIROWIT	DDDSOFTEOE
7	WHOFEETTT	SSEYGEMIEFTANED

Fig. 1-4, W

Note the steganography in the columns:

C24: DREWIED

DREW I ED

Jim and Ed Drew (masks)

C20: DAP

C19: ASOADO

C18: ILLMNSE

C17: ANSAIDI

DAP A, SO A DO FILL MN SE ANS AID I

Instructions: A pad (reversed), so A do fill in, See answer, Aid I. They used a "One Time Pad" with a 'A' as its letter mask. "SO A DO FILL IN"; Fill in the mask with K4 ciphertext,

K4-CT, i.e use pad letters as a key for K4-CT. See the answer, and aid Jim. The pad is reversed, which we did already in the step above. (Note that I also refer to this as the K4-HX pad)

C16: **CNETADM**

C15: **AREHEDE**

ARE H ED EM D AT ENC

Note a switch in the direction of reading; C15 forward then C16 backward.

Our H, Ed and Me, D at Encrypt. There is another mask 'H', start encryption at D; A clue to A Riddle X.

C02: **OEH**

C01: **LTYSAHW**

W HAS Y T L, HE O

W has why T Lied, He is O (T is O).

As you might notice, there is more in there but it easier to see when one understands more.

And another clue, when part of it is decrypted with 'V':

DEC(V, DREWIED) = HOIRNIH -> (backwards) H IN R IO H

H in our XH, There are two letters using the same pad, a H and an HX.

Main decryptions have steganographic clues along the edges of the masks indicating what to draw for the mask and giving other important information. In K4, the letters H, HX (also the A pad as discussed above, formed by two A's laying on their sides, i.e. the negative space around the H), O, P, and Y are the basic masks that I have found.

Part 1a: K4-HX

Transpositions Step:

Take the rows of W and arrange them as follows:

R5:R1

R6:R2

R7:R3

and as shown below in W1, Fig 1-5:

	1								2								1								2																							
	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1	2	3	4														
R5:R1	A	O	R	E	A	C	A	E	T	S	M	B	E	L	E	A	I	N	D	E	P	R	A	I	L	L	S	E	I	P	W	L	U	Y	E	T	H	A	C	A	I	A	D	E	S	B	D	
R6:R2	H	E	L	Y	O	I	E	T	I	R	O	W	I	T	D	D	D	S	O	F	T	E	O	E	T	M	W	M	L	K	R	I	I	P	L	L	D	Y	R	N	N	L	S	A	E	O	G	R
R7:R3	W	H	O	F	E	E	T	T	S	S	E	Y	G	E	M	I	E	F	T	A	N	E	D	Y	E	T	D	V	T	R	A	H	N	E	L	N	H	E	E	S	L	O	P	I	R	M	E	

Fig. 1-5, W1

Steganography of K4 HX Pad Mask:

Again, notice the clues in the steganography in Row 1, Columns 1 to 8 of W1, as shown in Fig. 1-5, spell out 'L WPIES LL'. Maybe it means 'L SPIES LL'; spies surrounded by L's. Also notice the last column of R5-R7 and the first column of R1-R3 spell out 'Y T LIED'. Last, notice that the first columns of R1-R3 and R5-R7 spell out 'DREW H, A' - so draw an H! (and some A's).

	1								2								1								2																							
	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1	2	3	4														
R5:R1	A	O	R	E	A	C	A	E	T	S	M	B	E	L	E	A	I	N	D	E	P	R	A	I	L	L	S	E	I	P	W	L	U	Y	E	T	H	A	C	A	I	A	D	E	S	B	D	
R6:R2	H	E	L	Y	O	I	E	T	I	R	O	W	I	T	D	D	D	S	O	F	T	E	O	E	T	M	W	M	L	K	R	I	I	P	L	L	D	Y	R	N	N	L	S	A	E	O	G	R
R7:R3	W	H	O	F	E	E	T	T	S	S	E	Y	G	E	M	I	E	F	T	A	N	E	D	Y	E	T	D	V	T	R	A	H	N	E	L	N	H	E	E	S	L	O	P	I	R	M	E	

Fig. 1-5b, W1 with H drawn on it

OTP, One Time Pad, Decryption Step:

The next step involves treating W1 as a pad and K4 as a ciphertext as part of a OTP, one time pad, encryption scheme. In this scheme, one uses the pad mask as a 'key' matched with parts of K4 as the ciphertext to recover the plaintext using the Kryptos Vigenère cipher, as was used in K1 and K2.

For each of the letters, I have tried to adopt a similar way of showing how to decrypt. First I show the letter pad mask, highlighted in colors, as in Fig 1-6, the HX Pad Mask. In the next figure, I show the K4 ciphertext separated out into rows and colored the same as the parts of the pad mask, as in Fig 1-7. The colored part of the pad mask form the keys used to decrypt the the same colored part of the ciphertext. Next I show them all together, the keys (pad mask), followed by the corresponding ciphertext, and last the plaintext found after decryption. The HX version of this can be seen in Fig. 1-8. I have laid it out such that one can easily use an online tool such as Rumkin's mentioned above. For that specific tool, one should set to Decrypt or Encrypt, depending on which one is doing. Sometimes one needs to Encrypt to get the plaintext used. In such cases I have denoted the font in blue instead of black. Then be sure to set the Alphabet Key to 'KRYPTOS', and copy and paste the keys into the 'Passphrase' box, and copy and paste the ciphertext into the 'Your message' box. Then one can read the decrypted message below.

[illegible]

Fig. 1-6, W1, HX Pad Mask

HX 1	OBKRUOXOGHULBSO
HX 2	LIFBBWFLRVQQPRNGKSSO
HX 3	TWTQSJQSSEKZZW
HX 4	ATJKLUDIAWINF
HX 5	NYPVTMTZFPKWGDKZXTJC
HX 6	DIGKUHUAUEKCAR

Fig. 1-7, K4 HX Ciphertext, HX-CT

	1	2	0	1	2
	123456789012345678901234	12345678901234	12345678901234	12345678901234	12345678901234
HX KEYS	OREACAETSMBELE ELYOIETIROWITDDDSOFT DIGKUHUAUEKCAR		UYETTHACAIADESB MLKRIIPLLDYRNNLSAEOG NELNHEESLOPIRM		
HX CT	ATJKLUDIAWINFB NYPVTTMZFPKWGDKZXTJC DIGKUHUAUEKCAR		OBKRUOXOGHULBSO LIFBBWFLRVQQPRNGKSSO TWTQSJSSEKZZW		
HX PT'	YPONBHZEROABUW BERLINHDEXPBCKJIMZTO VDRIDDLPIOIAQJ		DSIWLLLVSHZAWKW ZXFANBCKDFMNDBYANURM EFGRMOCKISWDXO		

Fig. 1-8

Now re-arrange by flipping the rows to get P1 as shown in Fig. 1-8,

DSIWLLLVSHZAWKW	YPONBHZEROABUW
ZXFANBCKDFMNDBYANURM	BERLINHDEXPBCKJIMZTO
EFGRMOCKISWDXO	VDRIDDLPIOIAQJ

Fig. 1-8, HX PT''

Misspellings Step:

Guess the misspellings as follows in Fig. 1-9.

Don't worry, there are ways to tell if you have the right ones. The rows labeled with 'HX MS' are the rows of the correct letters where a misspelling occurs in the plaintext, as best I can guess. The rows labeled with 'HX PT' are the plaintext rows from the OTP step. The rows labeled with W are the rows from the pad text and show the corresponding pad letter where the misspelling occurred in the plaintext.

[illegible]

Fig. 1-9

Which gives us HX-PT, our K4-HX solution to the 97 charters as shown in Fig. 1-10

DSISLLLVSOHAWKW YEONBHZEROABUZ
ZXFANBCQDFINDBYANORM BERLINHDEXPBCKJIMSTR
EFGSMARKISWDXO DDRIDDLEIOISQJ

or

D, SIS, LLL VS OH-AWK-WYE ON BH ZERO ABUZZ X
FAN B CQD FIND BY ANORM BERLIN HD EXP BCK JIM S
T REF GS MARK IS WD X
ODD RIDDLE IO IS Q J

Fig. 1-10, HX-PT

Examining the misspellings, Fig. 1-11, we can find further messages. Read the misspelled letters across, left to right, by rows going down in such a way to get the following, HX-PT, M1, as shown in Fig. 1-11.

Select the corresponding ciphertext from K4, as shown in Fig. 1-14.

H1 OBKRUOXOG
H2 HULBSOLIFBBWFLRVQQPRNGKS
H3 SOTWTQSJQ
H4 SSE
H5 KZZ
H6 WATJKLUDI
H7 AWINFBNYPVTTMZFPKWGDZXTJCDIGKUH
H8 UAUEKCAR

Fig. 1-14, K4 H Ciphertext, H-CT

Below is shown the K4 H pad, ciphertext, and resultant plaintext, Fig. 1-15:

		1		2		0		1		2																																																	
1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0
H KEYS																																																											
										WATJKLUDI										UAUEKCAR																																							
																				KZZSSE																																							
										OBKRUOXOG										SOTWTQSJQ																																							
H CT																																																											
										WATJKLUDI										UAUEKCAR																																							
																				KZZSSE																																							
										OBKRUOXOG										SOTWTQSJQ																																							
H PT'																																																											
										JMESIHQPK										TJIKESDD																																							
																				YHHIQH																																							
										GNIILXOQPT										QKGBJOCS																																							

Fig 1-15

Now re-arrange them. The left hand side of the 'H' reads bottom to top and is backwards, the right hand side is read top to bottom.

PQOXLIING	TJIKESDD
	HHY IQH
KPQHISEMJ	TQKGBJOCS

Fig. 1-16, H PT''

Misspellings Step:

Guess the misspellings as follows in Fig. 1-17.

The rows labeled with P1M are the rows of the correct letters where a misspelling occurs in the plaintext, as best I can guess. The rows labeled with P1 are the plaintext row from the OTP step. The rows labeled with W1 are the rows from the pad text and show the corresponding pad letter where the misspelling occurred in the plaintext.

	1	2	1	2
	123456789012345678901234	123456789012345678901234	123456789012345678901234	123456789012345678901234
H MS 1		L	O	C
H PT 1		JMESIHQPK	TJIKESDD	
W R5:R1	AOREACAETSMBELEA	INDEPRAI	LLSEIPW	LUYETTHACAIADESBD
H MS 1		W		
H PT 1		YHH	IQH	
W R6:R2	HELYOIETIROWITDD	DSOFTECE	TMWMLKRIIPLLDYR	NNLSAEQGR
P1M 3		Y	V	D
P1-R3		GNIILXOQP	TQKGBJOCS	
W R7:R3	WHOFEE	TTTSSEYGE	MIETANED	YETDVTRAHNELNHEESLOPIRME

Fig. 1-17

Examining the misspellings we can find further messages, reading going across:

LOC WY D,
Locate Why D

Which gives us H Plaintext, or H PT, our K4-H solution to the 97 characters as shown in Fig. 1-18:

PQO X LYING WHY KP Q HIS LM J
 TO IKES DCI Q H T Q KGB DOCS

Fig. 1-18, H PT

Part 2: K4-O

Transpositions Step:

As in part 1, start with W in Fig 1-3, but this time re-arrange the rows as follows:

R4:R3

R1:R6

R5:R7

shown below in W2, Fig 2-1, and draw the letter 'O' on it in the following way:

	1										2										0										1										2									
	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1	2	3	4																
R5:R3	A	O	E	A	C	A	E	T	S	M	B	E	L	E	A	I	N	D	E	P	R	A	I	Y	E	T	D	V	T	R	A	H	N	E	L	N	H	E	E	S	L	O	P	I	R	M	E			
R1:R6	L	L	S	E	I	P	W	L	U	Y	E	T	H	A	C	A	I	A	D	E	S	B	D	H	E	L	Y	O	I	E	T	I	R	O	W	I	T	D	D	D	S	O	F	T	E	O	E			
R4:R7	S	A	O	L	L	S	D	B	L	R	E	R	N	T	H	T	A	M	A	N	H	S	Y	W	W	H	O	F	E	E	T	T	S	S	E	Y	G	E	M	I	E	F	T	A	N	E	D			

Fig. 2-1, W2, O Pad Mask

Arrange K4 in as shown in K4 O-CT, Fig 2-2. Note that the ciphertext 'NYPVTTMZFPKWG', row O4b is repeated.

O1	O	B	K	R	U	O	X	O	G	H	L	B	S	O	L	I	F	B	B	W		
O2	F	L	R	V	Q	Q	P	R	N	G	K	S	S	O	T	W	T	Q	S	J	Q	
O3	S	S	E	K	Z	Z	W	A	T	J	K	L	U	D	I	A	W	I	N	F	B	N
O4a	Y	P	V	T	T	M	Z	F	P	K	W	G	D									
O5	K	Z	X	T	J	C	D	I	G	K	U	H	U	A	E	K	C	A	R			
O4b	N	Y	P	V	T	T	M	Z	F	P	K	W	G									

Fig. 2-2, K4 O-CT

Decryption Step:

As before in part 1, do the OTP decryption as follows. Note that the keys from rows R3 and R6 with the CT from rows O2 and O4b actually encryptions, with font in blue:

	1	2	0	1	2					
	123456789012345678901234	12345678901234	12345678901234	12345678901234	12345678901234					
O KEYS	EACAETSMBELEA INDEPRAI YETDVTRAHNELNHEESLOPI									
	LSEIPWLUYETTH WITDDDSOFTEOE									
	OLLSDLRLRERNTH TAMANHSYWWHOFEE TTTSSEYGEMIEFT									
O CT	OBKRUOXOGHULBSOLIFBBW FLRVQQPRNGKSSOTWTQJSJQ									
	YPVTTMZFPKWGD NYPVTTMZFPKWG									
	SSEKZZWATJKLUDIAWINFBN KZXTJCDIGKUHUAUEKCAR									
O PT'	QRLQDRMGOPTSRLFATCARB HYOSJXTBASEWZNIBDEENC									
	EWEIRUBLRINCV JLASHHXTXAEYX									
	RIQQILSSNIAGASBIJVOSV PENMOXSECQIPNQDNEXUW									

Misspellings Step:

Guess the misspellings as follows in Fig 2-4. The rows labeled with O MS are the rows of the correct letters where a misspelling occurs in the plaintext, as best I can guess. The rows labeled with O PT are the plaintext row from the OTP step above. The rows labeled with W are the row from the pad text and shows the corresponding pad letter where the misspelling occurred in the plaintext.

	1	2	1	2
	123456789012345678901234	12345678901234	12345678901234	12345678901234
O MS 1	A L O O ET S I			
O PT 1	QRLQDRMGOPTSRLFATCARBHYOSJXTBASEWZNIBDEENC			
W R5:R3	AOREACAETSMBELEA INDEPRAI YETDVTRAHNELNHEESLOPI IRME			
O MS 2	D C E T P K			
O PT 1	EWEIRUBLRINCV JLASHHXTXAEYX			
W R1:R6	LSEIPWLUYETTHACAIADESBDHELYOIETIRO WITDDDSOFTEOE			
O MS 3	R I M O E HY Q			
O PT 1	RIQQILSSNIAGASBIJVOSV PENMOXSECQIPNQDNEXUW			
W R4:R7	SAOLLSDLRLRERNTH TAMANHSYWWHOFEE TTTSSEYGEMIEFTANED			

Fig. 2-4

Read off the misspelled letters of Fig 2-4 starting by going up on odd rows, down on even, right to left for the right set of 24 columns and continuing then to the left set of 24 columns by going down on odd rows, up on even, to get the following,

KI**Q** P, **T**YH SET, OO MOLE CIA R D

K**I**M P, **O**YH SET, OO MOLE CIA R D

Kim Philby, OYH [are a] SET, OO [is a] MOLE, CIA ouR D

Information: K and P is for Kim Philby who is our double O mole.

The letters, OYH, are a set, and the CIA is D, i.e. the letter D stands for the CIA (possibly the CIA under Dulles). CIA RD can also mean CIA Riddle or 'OO MOLE [in the] CIA, Read'.

And following across, the letters of the pad where those misspellings occur we can get:

IN H D **A****E**S, SET D'S T, D R **A****F**S Y G T

IN H D **A****N**S, SET D'S T, D R **A****N**S Y G T

We get the following K4 O Plaintext, O-PT. Note that all but the last line of O-PTb should be reversed:

K4 O-PTa:

BR ACT OF LOST LOG A RD Q

LRQ V ENCR L BURIED E VS SOV

JIM S AGAIN IS LRQ Q

I RQ U X ENDY, H PIECES X O ONE P

K4 O-PTb:

I NEED B IN S WEST B TX JS O Y H

J LAST H, P TX KEY.

Steganography of K4 Y Pad:

Notice the letters in the center, reversed:

ORITE IO Y L **E**HD B'S E **D**IA C A

WRITE IO Y L **B**HD B'S E **C**IA C A

Write X, why Lie Berlin Head, B's E, CIA see answer.

Misspellings: O WEB D C

And also if decrypted with the key 'T'

Dec (T, ACAIADESBD HELYOIETIRO) = POPEPSAYTS DAGXREAKEWR

P - O PEP SAY T, S DAG X REAKE WR
P - O PEN SAY T, S DIG X READ E WR

As for the outside edges; on the right:
R, ME AN ED and the left: ALSO A R

So, the two different sections are drawn like R's.

Part 3: K4-Y

Transpositions Step:

As in part 1, start with W in Fig 1-3, and arrange the rows as follows:

R4:R5

R2:R6

R3:R7

shown below in W3, Fig 3-1, and draw the letter 'Y' on it in the following way:

1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4
 R4:R5 SAOLLSDBLRE **RNTH**TAMANHSY**WA**OREACAE **TSM**BELEA**INDE**PRAI
 R2:R6 TMWML **KRI**I**PL**LDYRNNLSAEOG**RH**ELYO **IET**IROWITDDDSOFTEOE
 R3:R7 YETDV **TRA**HNELN**NHEE**SLO**PI**RM**EW**HOFEETTSSEY**GEM**IEFTANED

Fig. 3-1, W3, Y Pad Mask

Y1 OBKRUOXOGHULB
Y2 SOLIFBBWFLRVQ
Y3 QPRNGKS
Y4 SOTWTQSJQSSEKZZWATJ
Y5 KLUDIAWINFBNYP
Y6 VTTMZFPKW
Y7 GDKZXTJCDIGKUHUAUEKCAR

Fig. 3-2, K4-Y Ciphertext

Decryption, and Encryption Step:

Done as before, please note that the left side, in blue should be encrypted:

		1		2		0		1		2														
	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1	2	3	4
Y KEYS																								
Y CT																								
Y PT'																								

Fig. 3-3

Misspellings Step:

Guess the misspellings as follows in Fig 3-4. The rows labeled with Y MS are the rows of the correct letters where a misspelling occurs in the plaintext, as best I can guess. The rows labeled with Y PT are the plaintext row from the decryption step. The rows labeled with W are the rows from the pad text and shows the corresponding pad letter where the misspelling occurred in the plaintext.

		1		2		1		2																
	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1	2	3	4
Y MS 1																								
Y PT 1																								
W R5:R3																								
Y MS 2																								
Y PT 2																								
W R1:R6																								
Y MS 3																								
Y PT 3																								
W R4:R7																								

Fig. 3-4, Y-PT Misspellings

To see the misspellings, read off the letters going down on even columns then up on odd ones, right to left:

WH **X** K **W**S L, **S**IA T - SAW T A I**B** O Q K HAD
 WH **A** K **I**S L, **C**IA T - SAW T A I**I** O Q K HAD

*White House Answer - Kim is Lie (is lying about) CIA T (Penkovsky, or YOGA)
 Saw T (Penkovsky, or true) Answer II (two - i.e. a double agent)? Kim Had*

Further misspellings: **I** **CIA** - **B** **SW** **X**:

*One CIA (One is to be reversed - E NO or expose not) - B (Blunt or Blake) SW X:
 perhaps a clue to the rocks.*

The pad letters where the misspellings occur give if read off going across left to right starting at the top, and then right to left for the next row, followed by left to right for the bottom row, give:

TTN'S A P T **L**R LI**K** **N**NS I E WOE'S G **E** A
TEN'S A P T **O**R LI**E** **A**NS I E WOE'S G **X** A

Misspellings: **T-L K N E** - **E O**, **E AX** *True or Lie, Kim Not Expose - Expose Oleg, Expose AX*

We get the following K4-Y and K4-N Plaintext:

K4 Y-PT:

V SPY T, JFK TNX - BH Y ANS P X
 A SENT V AG INFO SLAIN WD IS X
 HOW B

K4 N-PT:

D - SEE HOW Q
 B LL ALT Y - AB VS D, CIA X
 K III BHD T A X
 OP II, K TWO - AX L

Steganography of K4 Y Pad:

Steganography of the edges of the decrypting part of the letter 'Y'. Read off the letters going down then up, left to right:

R5 AOREACAE**T**SMBELEA**I**NDEPR**A**I
 R4 HELYO**I**ETIROWIT**T**DDDSOFTE**O**E
 R7 **W**HOFEET**T**TSSEYGEMIE**F**TAN**E**D

A HE OR **L****E** A, O C - A E, **D**D D'S OF-TE**A** **O**E, I ED
 A HE OR **L****E** A, O C - A E, **A**D D'S OF-T**E**N **Y**E, I ED

Answer of He or Lie Answer, O see - answer [should be] E, Allen Dulles and CIA [is] OF-TEN YE, I [and] ED . The CIA is often ye, and the CIA is of Ten (of the shadows)

Misspellings: **YDAO** - **IANY**:

Y D AOI ANY

Why D - AX Answer why; or A N/Y - Answer? No/Yes

Steganography of the edges of the letter 'N'. Read off the letters going up then down, left to right:

R4 SAOLLSDBLRE **R**NTH TAMANHSYW

R2 TMWML **K**R I I P L L D Y R N N L **S**AEOGR

R3 YETDV **T**RAHNELNHEESLOPIRME

YT SAME TWO LM D V, L LSD BL**R**E, **S**A**E** OGR

NT SAME TWO LM D V, L LSD BL**U**E, **S**E**E** OGR

Not [the] SAME, [the] TWO LM, D and V's, Lie LSD, BLUE - SEE O GR.

Not the same are the two LM's, CIA and the Cambridge V's Lie and LSD (and LL SD) BLUE (sad - or B Lie You Expose) . SEE OGR (a clue to one of the next letter masks, the G (OGR). LSD: Lysergic Acid Diethylamide, or Acid.

Further misspellings: **YRA**

NUE

N/Y RUE A

Yes/No Rue the Answer?

Steganography of the K4 Y Ciphertext:

Note that column Y3 is not used. Reversed it says:

S K**G****N** R P Q

A K**G****B** R P Q, *Answer - KGB our P? (Our Ye, since 'Q YE P'), the 'N' misspelling suggests no.*

Part 4: K4-P

This letter is a bit strange, it starts off as a 'P' and ends up as the letter 'E'.

Transpositions Step:

As in part 1, start with W in Fig 1-4, but this time re-arrange the rows as follows:

R2:R6
R5:R3
R1:R7

shown below in W3, Fig 4-1, and draw the letter 'P' on it in the following way:

	1	2	0	1	2
	123456789012345678901234	123456789012345678901234	123456789012345678901234	123456789012345678901234	12345678901234
R2:R6	TMWMLKRII	PLLDYRNNLSAEOGR	HELYOIETIROWITDD	DSOFTEOE	
R5:R3	AOREACAETSM	BELEA	INDEPRAIYE	TDVTRAHNELNHEES	LOPIRME
R4:R7	SAOLLSD	BLRERNHTAM	ANHSYWWHO	FEETTSSEY	GEMIEFTANED

Fig. 4-1, W4, P Pad Mask

P1 OBKRUOX
P2 OGHULBSOLIFBBW
P3 FLRVQQPRNGKSSOTWTQS
P4 JQSSEKZZ
P5 WATJKLU
P6 DIAWINFBNYPVTTM
P7 ZFPKWGDZXTJCDIGKUHUAUEK
P8 CAR

P9 FLRVQQPRNGKS

Fig. 4-2p, K4 P Ciphertext, P-CT

1 2 0 1 2

123456789012345678901234**41**23456789012345678901234

P KEYS

PLLDYRNNLSAEOGR**R**HELYOIETIROWITDDDSOFTEOE
TDVTRAHN ESLOPIR
FEETTSSEYGEMIEFTAN

P-CT

DIAWINFBNYPVTT**MZ**FPKWGDKZXTJCDIGKUHAUEK
JQSSEKZZ WATJKLU
FLRVQQPRNGKSSOTWTQS

P-PT'

AXJGGMNIYVVEZLL**LE**RFXXMXZVDWZNQSOPJICCPDSI
FDDYDNES FRGEWYQ
KSJMJJWUBEGUHJNEKGG

Misspellings Step:

Guess the misspellings as follows in Fig 4-4. The rows labeled with P MS are the rows of the correct letters where a misspelling occurs in the plaintext, as best I can guess. The rows labeled with P PT are the plaintext row from the decryption step. The rows labeled with W are the rows from the pad text and shows the corresponding pad letter where the misspelling occurred in the plaintext.

1 2 0 1 2

12345678901234567890123456789012345678901234

P MS 1 E S N I A L S S I A X K

P PT 1 AXJGGMNIYVVEZLLERFXMXZVDWZNQSOPJICCPDSI

W R2:R6 TMWMLKRIIPLLDYRNNLSAEOGRHELYOIETIROWITDDDSOFTEOE

P MS 2 J T X S O U S

P PT 2 SXUTXNDECWWZNT FDDYDNES FRGEWYQ

W R5:R3 AOEAECAETSMBELEA INDEPRAIYETDVTRAHNELNHEESLOPIRME

P MS 3 X N T I I G Y A

P PT 3 QBYAUGAIWQM KSJMJJWUBEGUHJNEKGG

W R4:R7 SAOLLSDBLRERNTHTAMANHSYWWHOFETTTSSEYGEMIEFTANED

Fig. 4-4

The misspellings of the decrypted part of the letter 'E' with the letter 'P'. Read off the letters going on the top row going across right to left, then the second row going left to right, and the third row going right to left again :

K X A IS SLAIN, SEJT X SO U SAY G - II, TNX
K X A IS SLAIN, SENT X SO K SAY G - II, TNX

? Kim X Answer is [Penkovsky] Slain, [Kim] Sent X - So Kim Say [to] G - II (that Penkovsky was a double agent), Thanks [for your info].

The pad letters where the misspellings occur give when read off going down on odd columns then up on even ones, left to right:

E TELL Y, T M,A OR T V, IEE T SIED E F IEN
E TELL Y, T G,A OR T V, SEE T LIED E F TEN

Expose - Tell Why ,True Golitsyn and Angleton or True [Cambridge] Five, See T [Penkovsky] Lied, Expose False Ten (Shadows)

We get the following K4 P Plaintext, K4 P-PT:

K4 P-PT:

AG KEY J HUGE BUG JIM IS K Q
S,W E GRU SEND Y O D, SIS KP X
CIA POS IN S WD VS LM X
F REAL LI-ENVY IN MSG E X A

To T and X Thanks, Jim.
And in the original line: ED NXT; thanks Ed.

E PT:

Q BY X U GAIN Q T
TNX WW C E TNX T U JS

Steganography of K4 P Pad:

Steganography of the edges of the 'P' letter. Read off the letters to get the following

R6 HELYOIETIROWITDDDSOFTEOE
R3 YETDVTRAHNELNHEESLOPIRME
R7 WHOFEETTTTSSEYGEMIEFTANED

WHO YE - E L NH E - ME, ED

WHO YE - E L BH E - ME, ED

Who Ye, Expose Lie Berlin Head [or] Expose Me and Ed

Steganography of the edges of the encrypting part, the 'E' letter. Read off the letters on the left side going upwards, right to left, then left to right, and then right to left:

R2:R6 TMWMLKRIIPLLDYRNNLSAEOGR
R5:R3 AO REACAETSMBELEA INDEPRAI
R4:R7 SAOLLSDBLRERNHTAMANH SYW

D'S L LOAS A OII R K LM, WM T

D'S L LOSS A III R K LM, LM T

D'S Lie (CIA's Lie), LOSS Answer III (triple) ouR Kim LM , LM True

Misspellings: A O W L IS: Answer Oleg - W Lie Is

And on the right side going upwards from left to right, the right to left:

A NH'S Y WIAR P, ED N I

A BH'S Y LIAR P, ED N I

Answer Berlin Head's (Allen Dulles) why, Liar P, Ed and I

Misspellings: NWBL: NW B L

Steganography of P CT:

Observe the two pieces, from Fig. 4-2 the P-CT or E-CT, that are not used: P1 and P8.

P1 OBKRUOX

P8 CAR

$\text{Enc}(T, \text{CAR}) = \text{GEO}$ and

X OUR K BOCAR

X OUR K BEHAR

Our Kim is Behar, with the misspellings forming the message C O HE, or: He saw O

Steganography of K4 CT:

1 ?OBKRUOX
2 OGHULBS
3 OLIFBBW
4 FLRVQQP
5 RNGKSSO
6 TWTQ SJQ
7 SSEKZZW
8 ATJKLUD
9 IAWINF B
10 NYPVTTM
11 ZFPKWGD
12 KZXTJCD
13 IGKU HUA
14 UEKCAR

Along the left side going down:

OO OF R T SAIN Z KI U

OO OF R T SAID Z KIM

And along the right side going down:

X SW POQ WD, B MDD A

X SW POS WD, B MAD A

And along the top:

?O B K RUO X, reversed X OUR K B O ?

E O B, K, RUO X

Expose Oleg, Blake, Kim, and Our Shadows, X

Also, take K4 H-CT, and add to it the two extra pieces of CT used in O-CT, O4b, and in P-CT, P9.

1	OBKRUOX
2	OGHULBS
3	OLIFBBW
4	FLRVQQP
5	RNGKSSO
6	TWTQSSJQ
7	SSEKZZW
8	ATJKLUD
9	LAWINFB
10	NYPVTTM
11	ZFPKWGD
12	KZXTJCD
13	IGKUHUA
14	UEKCAR

Fig. 4-3e

Reading off the unused letters by row from bottom to top, alternating from right to left and then left to right we find:

HUKGI
KZXTJCD
D
AWINFB
S
OLIFBBW
SBLUH

Guess the following misspellings and read off the correct letters the same way (left to right, right to left but going down)

H**G**K**B**
K**E**X**T**O**C**D
R
AWIN**G**B
S
OLIF**E****B****H**
SBLU**E**

Misspellings: **B GEORGE HE**

H, **G**, K**B** **KEX** **TO** C **DRAWING** B'S O **LIFE**, **BH'S** **BLUE**

Note: 'WHO YE' is a reference to P - the mask used:

Part 5: K4-G

Transpositions Step:

R1:R6

R2:R4

R5:R3

1 2 0 1 2

123456789012345678901234**1**23456789012345678901234

R3:R2 **Y**ETDVTRAHNELNHEESLOPIRM**E**TMWMLKRIIPLLDYRNNLSAE**OGR**

R5:R1 **A**O**R**E**A**C**A**E**T**S**M**B**E**L**E**A**I**N**D**E**P**R**A**I**L**L**S**E**I**P**W**L**U**Y**E**T**T**H**A**C**A**I**A**D**E**S**B**D

R4:R6 SAOLLSDBLRERNHTAMANHSHY**W**H**E**LYOIETIROWITDDDSOFT**E**O**E**

G1 OBKRUOXOG
G2 HULBSOLIFBBW
G3 FLRVQQPRNGKS
G4 SOTWTQSJSSEKZZW
G5 ATJKLUDIAWINFBNYPVTTM
G6 ZFPKWG
G7 DKZXTJCDIGKUHUAUEKCAR

Fig. 5-2, K4 G Ciphertext, G-CT

OTP Decryption Step:

123456789012345678901234**41**23456789012345678901234

G KEYS

	HNELNHEESLOPIRME	ETMWMLKRIIPLLDYRNNLSAE
AOREACAETSMB		LLSEIPWLU
	LRERNTHTAMAN	HELYOIETIROWITDDDSOFT

G-CT

	SOTWTQSJQSSEKZZW	ATJKLUDIAWINFBNYPVTTM
FLRVQQPRNGKS		OBKRUOXOG
	HULBSOLIFBBW	DKZXTJCDIGKUHUAUEKCAR

G-PT'

	MFNSESUOHIRBEXAF	PFNBKUCKMQXYYSMCDOWA
OFKEGEVJIABX		VZSFDBUVB
	WQSAGRPEOJRT	XEJKCOQHTHOMPZLOUSHNO

Fig. 5-3, K4-G

Misspellings Step:

Guess the misspellings as follows in Fig 5-4. The rows labeled with G MS are the rows of the correct letters where a misspelling occurs in the plaintext, as best I can guess. The rows labeled with G PT are the plaintext row from the decryption step. The rows labeled with W are the rows from the pad text and shows the corresponding pad letter where the misspelling occurred in the plaintext.

1 2 1 2

123456789012345678901234**41**23456789012345678901234

Y MS 1 KD WH T O V G D S

Y PT 1 MFNSESUOHIRBEXAFPFNBKUCKMQXYYSMCDOWA

W R3:R2 YETDVTRAHNELNHEESLOPIRMEETMWMLKRIIPLLDYRNNLSAE OGR

Y MS 2 P O D A L D E

Y PT 2 OFKEGEVJIABX VZSFDBUVB SVHSOW

W R5:R1 AOREACAETSMBELEAINDEPRAILLSEIPWLUYETTHACAI ADESBD

Y MS 3 A S U D H O

Y PT 3 WQSAGRPEOJRT XEJKCOQHTHOMPZLOUSHNO

W R4:R6 SAOLLSDBLRERNTHTAMANHSYWHELYOIETIROWITDDDSOFTEOE

Fig. 5-4, G-PT Misspellings

The G's misspellings, going across right to left then left to right, bottom to top:

OH D U SAP O DALDE, S D G VO T, H WD K
OH D U SAY O FALSE, S D G VS T, X WD K

Pad letters, going up and down, the down and up, right to left:

DEAO, D NT DRIW E LM ROLR N HEEO

DEAD, D NT DRAW E LM ROLE N HERO

DEAD (O), D NoT DRaW P, Expose LM ouR O, Lie Expose ouR HERO

G_a PT:

T R JOE, PR G, A SAW X
B AID V EGO KP, O FATE, BH WHOUSE, SND K AS X
ODD MSG Y X Q
V K, C UK B - NO P

G_b PT:

B V L B D F, SAVE O S D VS O-NOS HOLD P MOUTH Q O C K JS X

Steganography of K4 G Pad:

1 2 0 1 2
123456789012345678901234123456789012345678901234

R1:R6 YETDVTRAHNELNHEESLOPIRME TMWMLKRIIPLLDYRNNLSAE OGR
R2:R4 AOREACAETSME ELEAINDEPRAI LLSEIPWLU YETTHACAI ADESBD
R5:R3 SAOLLSDB LRERNTHTAMAN HSYW HELYOIETIROWITDDDSOFTEOE

ART V D, T E Y, SAO L LSD B
ART V D, T E Y, SAY L LSD B

E L EAIN, D E PR A I WYSH
E L VAIN, D E PR A I WISH

YE T THA C A I
YE T THX C A I

O GREO E
O GREY E

Part 6: K4 - X

This letter is the letter 'X'.

Transpositions Step:

As in part 1, start with W in Fig 1-4, but this time re-arrange the rows as follows:

R1:R7

R4:R3

R6:R2

shown below in W3, Fig 6-1, and draw the letter 'X' on it in the following way:

1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4

R1:R7 LLS EIPWLUYETTHACAIAD ESB DWHOFEE TTS SSEY GEMIEFTAN E D

R4:R3 SAOLL SDB L RERN THTAMANH SYW YETD VTRAHNELNHEE SLOPIRME

R6:R2 HELY OIETIROWITDDDSOFTEO E T M W M L K R I I P L L D Y R N N L S A E O G R

Fig. 6-1, W6, X Pad Mask

X1	OBKRUOXOGHULB
X2	SOLIFBBWFLRVQQPR
X3	NGKSSOTWTQSQJSSEK
X4	ZZW
X5	AT
X6	JKLUDIAWINFBNYPV
X7	TTMZFPKWGDZXTJC
X8	DIGKUHUAUEKCAR

Fig. 6-2, K4 X Ciphertext, X-CT

OTP Decryption Step:

123456789012345678901234**41**23456789012345678901234

X KEYS

EIPWLUYETTHACAIA TTTSSSEYGEMIEFTAN
 LRERNHTAMANHSYWYETDVTRAHNELNHEE
 YOIETIROWITDDDSOF KRIIPLLDYRNNLSAE

X-CT

ZZWOBKRUOXOGHULB JKLUDIAWINFBNYPV
 SOLIFBBWFLRVQQPR TTMZFPPKWGDKZXTJC
 NGKSSOTWTQSQSSEK ATDIGKUHUAUEKCAR

X-PT'

HDQBLOZDRQLSOHYR FVGITTODTRWWAXVP
 ITSHNTQNOZQPSHRT SIVCBARTRPEJLMRQ
 UMILDQOYRCDKTJFJF APUKDCCTNSYMC PKJ

Fig. 6-3, K4-X

Misspellings Step:

Guess the misspellings as follows in Fig 6-4. The rows labeled with X MS are the rows of the correct letters where a misspelling occurs in the plaintext, as best I can guess. The rows labeled with X PT are the plaintext row from the decryption step. The rows labeled with W are the rows from the pad text and shows the corresponding pad letter where the misspelling occurred in the plaintext.

1 2 1 2

123456789012345678901234**41**23456789012345678901234

X MS 1 S O D D A K

X PT 1 HDQBLOZDRQLSOHYR FVGITTODTRWWAXVP

W R1:R7 LLS EIPWLUYETTHACAIA DESBDWHOFEE TTTSSSEYGEMIEFTAN ED

X MS 2 NO P S O P G K

X PT 2 ITSHNTQNOZQPSHRT SIVCBARTRPEJLMRQ

W R4:R3 SAOLLSDBLRE RNTHTAMANHSYWYETDVTRAHNELNHEE SLOPIRME

X MS 3 YE O K X E X

X PT 3 UMILDQOYRCDKTJFJF APUKDCCTNSYMC PKJ

W R6:R2 HEL YOIETIROWITDDDSOF TEOETMWML KRIIPLLDYRNNLSAE OGR

Fig. 6-4, X-PT Misspellings

Misspellings reading up and then down, right to left by columns:

K X A, **K**D E G, X POS K OP DON**O**, E'S Y
K X A, **I**D E G, X POS K OP DONE**E**, E'S Y

Pad Misspellings:

NIE**H** Y E N R **H**, M Y V LEAD R D NO Y
DIED Y E N R **L**, M Y V LEAD R D DO Y

and

E Y HE IN **E**L**V**Y, **M**H N R Y, ODD R D A
K Y HE IN **E**N**V**Y, **B**H N R Y, ODD R D A

with misspellings: E LM K N B

X_a PT:

S D Q BLOOD RQ D'S O, H Y R F, V GIT TO D, DRAW A X V K, IT'S
NOT P NOS Q P SHOT! :(

X_b PT:

Q R KM G E PR TRAP **CV** IS F, JFK TO D CRY O Q D LIE Y

X_c PT:

A X UK D **C TEN SYM C** P X J

CV = 105 = onezerofive -> (reversed) E V, IF O REZ E NO
(expose V, if O is released, expose no)

C TEN SYM C = CXC = 190 = oneninezero -> (reversed) OR EZEN IN E
NO
(or even in exposure, no)

Steganography of K4 X Pad:

LLS**E**IPWLUYETTHACAIA**D**ESBD**W**HOFEE**T**TTSS**E**YGEMIEFTAN**E**D
SAOLL**S**DB**L**RERN**T**HTAMAN**H**SY**W**YETDVTRAHNELN**H**EE**S**LOPIR**M**E
HEL**Y**OIETIROWITDDDSOFTEO**E**TM**W**ML**K**RIIPLLDYRNNLSA**E**O**G**R

Far Left:

L'S **H**EAL SO L,L LSD B
X'S **D**EAL SO L,L LSD B

Middle Left:

ED BO**E**S T E D

ED BOSS T E D

Middle Right:

T WHMW O, FM LEE

T SHOW O, FM LIE

Far Right:

RED E MG O RIP, O L'S (as usual there are many possibilities)

RED E BG, O RIP, O T'S

Another example, Read'em (them) grey, O RIP, O T's (the truth of Penkovsky)

Part 6b: K4 - O

This letter is another 'O' with IIII, or four I's (meaning just FOUR?).

Transpositions Step:

As in part 1, start with W in Fig 1-4, but this time flip the the rows as shown:

R7:R1

R3:R4

R2:R6

shown below in W3, Fig 6-1, and use the negative space from the 'X' in the following way:

1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4

R7:R1 **WHO**FEE TTTTSSEY GEMIEFTAN **ED**LLS EIPWLUYETTHACAIA **DES**BD

R3:R4 YETDVTRAHNELNHEE **SLO**PIRME **ES**AOLLSDB LRERNTH TAMANHSYW

R2:R6 **TM**WML KRIIPLLDYRN NLSAE **OG**RHELYOIETIROWITDDDSOF **TE**OE

Fig. 6-5, W6, X-0 Pad Mask

And use the following as the ciphertext:

1	OBKRUOX
2	OGHULBS
3	OLIEBWB
4	FLRVQQP
5	RNGKSSO
6	TWTQSSJQ
7	SSEKZZW
8	ATJKLUD
9	IAWINFB
10	NYPVTTM
11	ZFPKWGD
12	KZXTJCD
13	IGKUHUA
14	UEKCAR

Fig. 4-3e

Quickly encrypting it:

1. Enc(**HULBSOLIF**, **DESBDEOE**) = XSWJJCYQW
2. Enc(**WHOFEE****TMWML**, **BWFLRV****QQPRN**) = OELPFAXFKND
3. Enc(**AWINFB****NYPVT****TM**, **OG****RSLOPI****RMEED**) = FDJZPGVLTHIIY
4. Enc(**FPKWGDKZXTJ****CDI**, **LLS****SAOLLSDB****HEL**) = PQSPQILJTHXWUS

Fig. 6-6, K4 X-O PT'

PT 1: X SW JS C YHW

X SW JS, see why

PT 2: **HELP** **T**, A X **FIND**

Help T (Penkovsky and Truth), Shadow Answer Find

PT 3: FD JS **PO** V L **THIS** Y

Find, JS, Penkovsky verses Lie, this is why

PT 4: **POS** P QIL J, **TNX** W US

*Position on Penkovsky? (a Lie) J, Thanks DCI Webster, us (Jim and Ed)???
(normally W can mean Allen W. Dulles - but this interpretation doesn't make sense to me)*

And the unused text from K4 CT? Reading up:

HUKGZSKGB -> **H**, UK, G **ZS** KGB
D, UK, **G** IS KGB

The CIA, UK (SIS) are Grey [and] KGB and also G is KGB i.e. G is used for short, KGB.

The Drawings:

Listing the letters we have found

A HA OR NYEP OR G X - O IIII which looks a lot like the following:

A **HA** OR NYE**P** OR G X - O **IIII**
A **DA** OR NYE**T** OR G X - O **F OUR**

'Da' would be yes in Russian, and 'Nyet' would be no, thus giving:

Answer 'Da' (yes) or 'Nyet' (no) or G (grey) X - O (Penkovsky) [is] OUR Fault

The Cast:

H, HAR, K, KP, KM:

*Harold Adrian Russell "Kim" Philby or HAR Kim Philby
Kim from KIMBALL O'HARA - Kipling's Kim*

B, GB, BG:

George Blake (born George Behar)

B

Anthony Blake

J, JIM, JIM S, JS:

Herbert James Sanborn Jr, or just Jim Sanborn

ED:

Ed Scheidt

D:

CIA

S:

SIS

BH:

Berlin Head, DCI Allen Welsh Dulles - or just a reference to a DCI.

G:

KGB or in one case Great Britain, or Anatole Golitsyn

O, P, T, PO, OP:

Oleg V. Penkovsky

CV or V:

Cambridge Five spies

V:

Also it can mean Verses

W:

Also Allen Welsh Dulles

WW: DCI William Webster

KM:

KIM PHILBY

L, C, Y, A and E

In most places L is for LIE, C is for SEE, Y is for WHY, A is for ANSWER and E is for EXPOSE - or if you wish "ENLIGHTEN"

And also keep in mind:

H JIM SANBORN, HAR "KIM" PHILBY, GEORGE "BEHAR" BLAKE and thus confusion.

Interpretations:

What follows are my best guesses to date - certainly not everything will be correct.
CQD! SOS!

Plaintexts and others:

HX PT :

D SIS LLL VS O-H-AWK WYE ON BH ZERO ABUZZ X
FAN B CQD FIND BY ANORM BERLIN HD EXP BCK JIM S
T REF, GS MARK IS WD X ODD RIDDLE IO IS Q J

D¹ SIS² LLL VS O-H-AWKward WYE³ ON BH ZERO⁴ ABUZZ⁵ X
FAN of B CQD⁶ - FIND BY AbNORMal BERLIN HeaD, EXPeCt BaCK - Jim S.
T is REFeRence⁷
GS MARK IS WD⁸ X
ODD RIDDLE - 10 IS?⁹ Jim

1. D is short for the CIA, or the CIA under DCI Allen Dulles.
2. SIS: www.sis.gov.uk , The British Secret Intelligence Service, aka MI6
3. LLL vs OHW: LLL, maybe Lies, Lies, Lies or, "Tri-L" for trial, or "try L", or LLL is a letter mask for LSD, i.e. Lies, Lies, Lies, are covering up LSD.

If 'LLL' reads 'Trial' or 'Try L' then we may have:

ENC(L, OHWYE) = V OHNY

Maybe that means to 'V' 'OHNY' so (remember 'Level Two'):

ENC(V, OHNY) = RD I X

Maybe that means that we should 'X Riddle One' i.e. encrypt K4-H with an X.

A. So the first line could be read as CIA, SIS LLL VS O H WYE ON BH, ZERO ABUZZ X

O a reference to Oleg Penkovsky,
H a reference to H.A.R. "Kim" Philby,
WYE ON BH: (the reason) Why is known by the Berlin Head, Allen Dulles
ZERO ABUZZ: Zero could stand for O or be misspelled, for Hero - both referring to Oleg Penkovsky.
Penkovsky is ABUZZ (sent a BUZZ or a telephone call).

Allen Dulles the was head of the OSS in Berlin just after WWII.

B. Or could be read as CIA, SIS LLL VS OHW (AWK W), YE ON BHZERO, A, BUGZ X

CIA and the SIS's Lies, Lies, Lies Vs an Awkward WHO.

'Ye' as part of John 8:32 'And Ye Shall Know ...' which is carved into a wall on the OHB (Old Headquarters Building) in Langley. In which case YE and the letter A, maybe in SHALL, are bugs, i.e. are wrong.

BHZero: OHB written backwards, Old Headquarters Building, Langley, CIA
For the clue on why BUGZ see the misspelling for K4-H below.

6. FAN B CDQ: A Fan of 'B', must mean fan of Anthony Blunt. CQD is an old form of SOS. It could also be meant to be read 'Seek D'.

7. T (for Truth) is our reference point referring to 'T IS YOUR POSITION' in Morse code next to the loadstone. The compass is pointing SW - (maybe SW is short for the SIS and W (Allen W. Dulles))

8. GS MARK IS WD X - on the USGS Marker that was in front of the NHB, New Headquarters Building. On it is written the word (letter) 'X' (or a cross, depending on your point of view)

9. Odd riddle: yes this is. Ten is?

ENC(V, ENDY) = AISX

The "Answer IS Shadows" - our Riddle

HX PT: M1

SO EZ Q IOS RSA R DES

SO Easy? aS RSA oR DES

Note that if one does the following:

ENC(AISX, SOEZQIOSRSARDES) = GQLWRTETBUGZLKF

G Q L, WRTE T, BUG Z, L K F
G ? Lie, and Write T; there is a Bug Z; L K F.

*Or maybe: George Q Lie, Write T, BUG Z (BUZZ should be written as BUGZ or Bugs),
Lie Kim False*

HX PT: M2

S W TO ZIE PRZ WEQK L IM YOUES Z O ROTS R NAOE RCED V HEP T SAY

Maybe:

SIS, Allen W Dulles T-O Lie, Mr President weak lie. I'm Yours, O Rots our name,
Received V Help Say the Truth.

Or better:

SIS and Allen Dulles's Lied about Penkovsky, Mr President, is a weak lie. I'm Yours, O (what we did concerning Penkovsky) rots our name. I received five documents and help. Please, the Truth say (Declassify the truth about Penkovsky).

O is for Oleg: Oleg Vladimirovich Penkovsky, a soviet spy for the US and Great Britain, and so is T. Oleg Penkovsky's SIS Code name was YOGA:

Dec (X, YOGA) = TIAC or backwards CIA T.

The CIA code name for him was HERO.

H_a PT:

TO IKE'S DCI Q H T Q KGB DOCS

Ike's DCI: again Allen Dulles.

Q H T Q:

Who is right? H? or T?

H, Kim Philby? or T, Oleg Penkovsky?

H_b PT:

PQO X LYING WHY, KP Q HIS LM J

PQO: OVP backwards: Oleg Vladimirovich Penkovsky.

KP: Kim Philby

LM: Lucid Memory

O_a PT:

BR ACT OF LOST LOG A RD Q

LRQ V ENCR L BURIED E VS SOV

JIM S AGAIN IS LRQ Q

I RQ U, X, ENDY, H PIECES X

O ONE P

BRother¹ ACT OF LOST LOG Answer Read? Location Request:

V ENCRypted L BURIED E VS SOVIets²

JIM S. AGAIN IS (this) a Location Request? I require you, a X, the ENDY, and H PIECES X O one P.³

1. Brother meaning the British, or just 'British'. The rest can read OF LOST LOG MAD? if encrypted with B, MAD deciphers to KIM. Or maybe its LOGA is misspelled and it should read: BR ACT OF LOST YOGA RD: Brother Act of Lost YOGA ReaD (or Riddle)? With YOGA being again a reference to Penkovsky.

2. V ENCR L BURIED E: suggests that there are five buried encrypted documents that Jim hid; L being for letters in this interpretation, and that they are documents that Expose (or Enlighten) our dealings verses the Soviets regarding the Penkovsky affair.

3. 'ENDY' is the ENDYAHROHN piece, 'H' meaning our PT-H, and a hint that the mask for this part forms an O and that O is the same as P (Oleg Penkovsky). And note the last 3 letters 'n' 'e' 'P' are clues to more letter masks. (i.e. that there should be a N and an E and a P).

O_b PT:

I NEED BINS WEST B TX JS O,Y,H

I NEED BINS WEST Berlin Text¹, Jim S. O-Y-H²

1. Binary or it might refer to JFK's "Ich bin Ein Berliner" speech.

Or Maybe it should read: I NEED B IN S-WEST B TX [I NEED B in South-WEST B TX]. Perhaps a reference to a location (T is your position ?) where something is buried, or a Key found there for decryption.

2. O-Y-H a clue to Oleg, YOGA, HERO - the letters of the masks of K4, along with the last 3 letter of PT-O a, N E P

O_c PT:

J LAST H PT X KEY X

Jim - LAST H Plaintext X KEY X

The last part of PT-H is key; meaning the riddle of 'IO IS ?' which is A IS X, and that is a Key. Also it is a hint as what to do with the PT-H plaintext (encrypt it with an X).

Y PT:

V SPY T, JFK TNX BH, Y ANS P X
A SENT V AG INFO, SLAIN WD IS X
HOW B

[Cambridge] V [say] SPY (Penkovsky) [is] True. JFK Thanks to BH (Berlin Head) [for] WHY ANSwer P X

Answer (or maybe Angleton) SENT V AG (A. Golitsyn) INFO, SLAIN WD IS X HOW B

Cambridge Five (Kim) say Penkovsky is a true double agent (not triple). JFK to Allen Dulles - Thanks. Why was the Answer an X (Penkovsky was executed). The Answer: (or Angleton) sent Kim (V) Anatole Golitsyn's information, and Kim was asked to send a message back as to the result of the operation to save Penkovsky - an 'X' if Penkovsky was executed. How did it fail - because of B (Anthony Blunt).

N PT:

D - SEE HOW Q
B LL ALT Y - AB VS D, CIA X
K III BHD T A X
OP II, K TWO - AX L

D SEE HOW Q
B LL ALternative whY - Anthony Blunt VS Dulles and CIA X
Kim III, Berlin Head True Answer X
Oleg Penkovsky II, K TWO - A is X is a Lie?

The other answer why: D - See How? Blunt Lie Lie
Alternate why - Anthony Blunt Vs Dulles and the CIA
Kim Philby was a triple agent thought Allen Dulles (True A is X or just True Answer).
Oleg Penkovsky was a double agent and so was Kim Philby - A is X is a Lie

P PT:

AG KEY J HUGE BUG JIM IS K Q
S, W E GRU SEND Y O D, SIS KP X
CIA POS IN S WD VS LM X
F REAL LI-ENVY IN MSG E X A

Anatole Golitsyn KEY J,
HUGE BUG JIM IS Kim Q
Sis, (Dick) White Expose GRU - SEND whY Oleg to D, SIS, KP X
CIA POSition in South WorD VS LM X
False REAL LI-ENVY IN MSG Expose X Answer

Anatole Golitsyn is key J, the huge bug, Jim, is Kim?
SIS and Dick White (the head of MI6)¹ [will] Expose GRU - Send why Penkovsky [lied]²
[to] the CIA, SIS and [give it to] Kim Philby.
See the CIA's position under a south word verses (Kim's) lucid memories.
They (lucid memories) are false the real reason is ENVY in his (Philby's) Message to
Expose that X (the shadows - CIA etc) is a Lie.²

1. Or maybe it could mean South West Expose, that the document that details this is located at a South West Position amongst the Morse Code on the rocks.

2. A Clue to where both items are buried. At the morse code on the rocks, the last T in INTERPRETATI.. which is due south of the compass rose hold the CIA position, i.e.. the Truth (I think). The L of Lucid Memory is where the Lie is buried; possibly something from Kim Philby.

E PT:

Q BY X U GAIN Q T
TNX WW C E TNX T U JS

? By X you gain ? T,
Thanks WW, see E
Thanks T and you, JS.

? By shadows you gain truth (or Penkovsky)?
Thanks DCI William Webster, see - enlightenment (exposure).
Thanks to Penkovsky and to you, Jim Sanborn.

E PT:

Q BY X U GAIN Q T
TNX WW C E TNX T U JS

? By X you gain ? T,
Thanks WW, see E
Thanks T and you, JS.

? By shadows you gain truth (or Penkovsky)?
Thanks DCI William Webster, see - enlightenment (exposure).
Thanks to Penkovsky and to you, Jim Sanborn.

G_a PT:

T R JOE, PR G, A SAW X
B AID V EGO KP, O FATE, BH WHOUSE, SND K AS X
ODD MSG Y X Q
V K C UK B - NO P

T ouR JOE, PR G Answer SAW X
B AID V EGO KP, O FATE, Berlin Head - White HOUSE.
SeND K AS X
ODD MSG Y X Q
V K See UK B - NO P

True our Joe [Bulik], a Public Relations (PR) campaign against G (KGB) [as the] answer saw. Blake aids [Cambridge] Five Ego [and E GO] Kim Philby i.e. to expose, goes Kim Philby [to Russia with O's fate in the balance]. Word of Penkovsky's fate to the DCI and White House sends Kim - the message is an X. An odd message, why X?
The Cambridge Five and Kim see the UK and see that they don't know about Blunt? So no Penkovsky (the US will not get Penkovsky).

G_b PT:

B V L D F, SAVE O S D VS O-NOS HOLD P MOUTH Q O C K JS X

Blake, Five Lie D False. SAVE O SIS, D Vs O-NOs HOLD P MOUTH? O See Kim JS X

Blake and the Cambridge Five's Lie is found to be false. The plan to save Penkovsky by the SIS and CIA versus Oleg's numbers (or oh-no's) hold Penkovsky's mouth (he doesn't divulge who his SIS and CIA handlers are) ? Oleg meets with Kim Philby.

X_a PT:

S D Q BLOOD RQ D'S O, H Y R F, V GIT TO D, DRAW A X V K, IT'S NOT P NOS Q P SHOT! :(

SIS D ? BLOOD RQ D'S O, H whY aRe F, V GIT TO D, DRAW A X V Kim, IT'S NOT P Numbers? P SHOT! :(

SIS and CIA, Blood required [since] the CIA's "O why" (LSD) and "H why" (about Kim) are false (or at fault), the [Cambridge] five (i.e. Kim) get to the CIA [a message], [they] draw a 'X' - It's not Penkovsky's numbers - which mean he's been shot! :(

Note: ten -> net ~ nyet or 'no' in Russian

V -> five -> evif ~ expose five if ... (you don't give us Penkovsky) ?

X_b PT:

Q R KM G E PR TRAP **CV** IS F, JFK TO D CRY O Q D LIE Y

? ouR Kim, KGB PR TRAP CV IS F, JFK TO D CRY O? D LIEd Y

? Our Kim Philby and KGB trap CV is False. JFK says to the CIA that he is upset about Oleg (being executed). Why did the CIA lie?

Note: **CV** = 105 = onezerofive -> (reversed) E V, IF O REZ E NO
(expose V, if O is released, expose no)

X_c PT:

A X UK D **C TEN SYM C** P X J

Answer X UK, CIA see TEN SYMbol see, Penkovsky X'ed (Executed), J

The answer 'X'. The UK and the CIA see a Ten's symbol, see Penkovsky X Jim.

Note: **C TEN SYM C** = CXC = 190 = oneninezero -> (reversed) OR EZEN
IN E NO (or even in exposure, no)

X-O PT:

See section 6b.

Opinion:

Bismillahi Al Haqq il Ala,
(In the Name of God, The Truth, The Most High)

"AND YE SHALL KNOW THE TRUTH AND
THE TRUTH SHALL MAKE YOU FREE."

Jesus the Son of Mary, the True Messiah, Alayhe Salam
John VIII - XXXII,
The Gospel

The Biblical and Islamic Exegesis of "Truth" in this verse is that God is The Truth, that He is the One True God, the only One deserving of worship.

However, I think Jim is using this verse to refer to a different truth. Perhaps a truth that has long been hidden. In his version, who is Ye? Is it Kim Philby or George Blake?; Or perhaps Allen Dulles, or the CIA. The BUGZ A, and YE would yield for example:

AND Allen Dulles, SIS, CIA, X LL KNOW THE TRUTH AND THE TRUTH SHALL MAKE YOU FREE.

In this case:

YE = Allen Dulles

S = SIS, MI6

H => D = CIA

A = X

YOU = YOGA, Oleg Penkovsky

Hopefully, the bugs will be fixed and YE will mean all of us before too long.

Kryptos is a memorial to Oleg Vladimirovich Penkovsky who's initials form four of the first five letter masks used upon Kryptos: HYOP

HERO: CIA Codename

YOGA: SIS Codename

OLEG

PENKOVSKY

Oleg V. Penkovsky, who spied for us from the heart of the Soviet Union. The information given to us, the United States, by Penkovsky was of the most critical importance. It assisted our President, John F. Kennedy, during the Cuban Missile Crisis in properly assessing the situation of Soviet nuclear forces in Cuba, and helped bring the crisis to an advantageous end for the United States. What Penkovsky did, at

enormous risk to himself, quite possibly prevented a nuclear holocaust. To that we are all in his debt.

I believe another point Kryptos tries to convey is the "Elusiveness of Truth". The world of spies is one of shadows, mirrors, and deception. There is no deciphering algorithm which can measure the truth content of a message. It serves as a warning and lesson to those working in the CIA and the rest of our intelligence community: Be careful with interpreting information, and be diligent. For how was Penkovsky found out? Penkovsky, as we can now guess, was implicated to the KGB by George Blake. and executed. Many of our spies were also lost to Kim Philby and his Cambridge Five. All of whom were working within our and our allies intelligence communities.

Intelligence, its gathering and analysis, is critical for our Nation. When it is not done well, or when it is not well heeded, all of us Americans suffer its consequences as does suffer our great Name and Liberty. But, when done right, and when acted upon with prudence and wisdom the results are magnificent; the results are like MAGIC. For how can we stop, or at least be prepared for the next war? How can a cold war be prevented from going hot? How can a nuclear holocaust be prevented? Why is there hope of any of this: Indeed as the anagram of the masks asks, Why is there a hope?: the Answer IS our ShadXws. There is hope due to good and dangerous work our spies, like Oleg V. Penkovsky, and the people of the CIA, and the Intelligence Community of the United States of America do which protects our Nation and its Liberty. And for that I thank them.

... And while this is undoubtably true, Kryptos seems to point to more whys. One of which is "why T lied", or why Penkovsky lied. Lied about what? Kryptos seems to imply that some injustice has been done concerning him - or perhaps, one guess might be, that there is some piece of information which would absolve him of this lie but for security reasons could not be revealed at the time, and still had not been at the time of Kryptos' creation.

There are many possibilities, as we can guess, since Penkovsky was found out by the KGB - but not immediately arrested it seems. I am hesitant to say the following since I have only done some basic reading, and I am certainly no expert in this field i.e. I do not know the subject matter well.

So with speculation, one guess might be what is talked about towards the end of an interview with one of his CIA case officers, Joe Bulik, given to "The National Security Archive" at The George Washington University:

<http://www2.gwu.edu/~nsarchiv/coldwar/interviews/episode-21/bulik1.html>

It concerns an agreed upon message that was only to be sent if the Soviet Union was starting a surprise nuclear attack on the United States and our allies:

Mr Bulik: *"... We had to have a contingency plan that if he were unable to get to anybody, and the biggest worry that the US military had was that if the Soviets planned a pre-emptive attack upon the United States by way of missiles, to give us a certain signal"*

And,

Mr Bulik: *"Well, when we working communications with him, besides the human contacts and the radio contacts, we had to have special signals from him in the case that he couldn't meet with anybody, but could, by way of telephone, give us the signal and it wasn't until we learned later that after he was captured that a signal came through that the Soviets were sending missiles over or planned to send missiles to the United States to attack us."*

Penkovsky has been accused by some that he tried to bring an end to the world - after trying to save it:

<http://www.thedailybeast.com/articles/2013/11/03/the-spy-who-saved-the-world-then-tried-to-destroy-it.html>

In other words, he lied to his interrogators and told them to send a message he knew which could lead to the starting of nuclear war. This might be the lie, and if this is what is being referred to, maybe Kryptos knows something else. Something that tells a different story. Why was the message sent, when it was not true. Why was Penkovsky abuzz? Since Penkovsky was in KGB hands when the telephone call was made, it stands to reason that the call must have been made by the KGB. It is presumed then that KGB must have found out about the message during their interrogations of Penkovsky.

Kryptos suggests that the reason why Penkovsky lied was due to the KGB's use of "enhanced interrogation techniques". Namely, that the KGB had drugged him with LSD. Possibly, they used LSD as a truth serum, or setup a situation that lead Penkovsky, while drugged in a hallucinated state, to truly believe that the Soviets were about to launch an attack and reveal the coded actions. It is clear that they used this signal as a trap to capture one of our Agents handling Penkovsky, Dick Jacobs.

But when the CIA found out what had happened, why did the CIA , at best, leave it to be suggested that Penkovsky had signaled the US and Britain that the Soviets where preparing to launch an attack and presumably did this to get the US to launch an attack against the Soviets?

Kryptos continues with the plan to save Penkovsky. Joe Bulik came up with a PR plan to save Penkovsky (again the same interview):

"Well, during the course of the trial, there's no question either at the beginning of the trial or the middle of the trial or the end of the trial, there was no question in my mind

that Oleg would be executed. And the first thought in my mind was, can I save his life? Can I save him from being executed? So I proposed a plan, a copy of which I still have today, I brought it with me, I stole something from the CIA, although they have copies of it. This is a plan that was to go to four KGB residencies in the world and four GRU residencies in the world and the reason I did that was to have the KGB and the GRU officers throughout the world know that the Americans were trying to save Oleg Penkovsky's life. And the idea was that we had enough information that was very embarrassing, not only to the KGB, but to the Soviet government, tremendously embarrassing to them, that if they executed him, that we would publish that information in the western press..."

So his plan was to embarrass the KGB, and Penkovsky wasn't telling all that he knew either, as Mr. Bulik further explained:

"He kept the identities of his old case officers secret and when the Soviets published a book on the Penkovsky case and the trial, they couldn't identify the four case officers, 'cos he figured we might possibly save his life and I tried to save his life."

However, another plan was hatched. The White House and the CIA tried to save Penkovsky, and it seems that they sent (or let go) Kim Philby to the Soviet Union. I'm not so clear as to the details of their plan.

To back up a bit, George Blake was finally exposed and caught by information provided by Michael Goleniewski, a Polish triple agent who defected to the CIA in January of 1961. In addition, other moles and Soviet agents were caught with the information that was provided by him. Later that year another KGB agent, Anatole Golitsyn, defected to the CIA in December. Golitsyn brought news of more moles within the western intelligence community. In specific about Britain, he told of the KGB's "Ring of Five" spies, but it seems that he did not know who they all were, although he is credited by many for providing the final proof that Kim Philby was one of them (however not by DCI Stansfield Turner²). *"Golitsyn's definition of the Ring of Five was taken so literally not merely by Wright [the head of MI5 at the time] and the small band of conspiracy theorists but by other Service investigators"*³. If that was the case, then they would have had four of them already: Maclean, Burgess, Philby, and Cairncross (he had confessed to MI5 years before). So they would have been looking for another one. All were rumored to have gone to Cambridge together, but if the British had not considered this fact, or didn't know it at the time, they might have included George Blake in the group, and thus considered that they had found all of them.

Maybe the plan went something like the following, and note that one must place themselves in the context of the early 1960's to understand what was happening. I believe, that Kryptos says that the operation was run by Allen Dulles. It seems, from the 'N' pad of the 'Y' letter pad, that Allen Dulles thought that Kim Philby was a triple agent, that he was really working for SIS and the CIA. It has been said that James J. Angleton, who was the head of the CIA's counterintelligence program, had also

thought the same thing. So they sent Kim Philby, with information from Golitsyn [again from the 'Y' letter pad], to the Soviet Union to find out about Penkovsky. Was Penkovsky really a triple agent as Golitsyn suggested? Why was the message sent that the Soviets were about to launch an attack? If so could they bargain with Joe Bulik's PR information for Penkovsky's life - threatening to expose the KGB and the GRU? They also might have told Kim Philby that Golitsyn had told them of the five spies and outed him being one of them.

Kim Philby in his book, "My Silent War", eludes with the following quote:

"So, after seven years, I left Beirut and turned up in the Soviet Union. Why? Maybe I was tipped off by a Fourth Man. Maybe someone had blundered. It is even possible that I was just tired."

Indeed, whatever the case someone did blunder - and it seems Philby knew that they did not know all five of them, and he had seen that Penkovsky was not a triple agent. So when Penkovsky was executed and when Kim decided to stay in Russia, as he had been a double agent all along, it must have sent the CIA and SIS into panic mode. Were they still penetrated, as Golitsyn suggested? Soon after, the "Forth Man" was found to be Anthony Blunt. Blunt was given immunity by the British for telling all that he knew, and his treason was covered up. Blunt was still allowed to continue his regular life, part of which was the "Surveyor of the Queen's Pictures" in addition to being eventually becoming a professor of art history at the University of London, and director of the Courtauld Institute of Art. He was also related to the Queen Mother.

Why was he given immunity, especially when a couple of years before, Blake was given 42 years in prison? I don't know but perhaps a new deal was struck up with the Soviets, one that didn't include saving Penkovsky. Perhaps it involved giving Blunt immunity. It is no exaggeration how embarrassing it would have been if the Soviet's reviled SIS and the CIA messed up by sending Philby or that their agents had penetrated from MI5 and MI6 all the way to the Royal Family. (if this is correct, it might be said that the Penkovsky affair was mostly covered up in part due to art - and now revealed due to art). A hint of this might be found in a paragraph from the book "Dead Drop" by Jeremy Duns:

"Two years later [1964?], when the Sunday Telegraph serialised Greville Wynne's [a British intelligence worker caught by the Soviets from his work in the Penkovsky case] account of his involvement in the operation, Wolfenden [a newspaper reporter who was working for the KGB and MI5, playing both sides] wrote an article claiming that the Soviets were considering releasing 'their own version of what Mr. Wynne said under interrogation'. The implication was that Mr. Wynne something to hide, and that if too much propaganda hay was made of his account the KGB would reveal some embarrassing secrets. They never followed up on this threat, but Wolfenden's article left the impression that crucial information was being omitted, and that the Russians had a hold on British intelligence"

In Kryptos, there are hints that Kim's Lucid Memories say that the reason why Penkovsky was not saved or cleared of lying was ENVY. That might be the case, but it also might be deceiving, for there is another interpretation of this word.

E N V Y: Expose Not [Cambridge] V - [that is] whY.

Also note:

If one decrypts the X_b plaintext with the key 'V' one might find an answer:

Dec(V, Q R KM G E PR TRAP CV IS F, JFK TO D CRY O Q D LIE Y) =
X O TV L I AO BOEA GK ND J, QJT BC H GOS C X H UNI S

which reversed reads:

SIN **U**, H X C SO G, H C B T J Q, J D **N** K GA**E** - O, B, **O**AIL V TO X
SIN **G**, H X C SO G, H C B T J Q, J D **A** K GA**Y** - O, B, **W**AIL V TO X

misspellings: U N E O, GAY, W

and doing the same with the X_c plaintext gives:

Dec(D C TEN SYM C P X J) = H, G, BI W D'S V GAY Q

If we look at the end of this text we see "W, D's V GAY ?"

The two together I believe, as unbelievable as it sounds, suggest that Allen Dulles and the CIA threatened to expose Kim Philby and the Cambridge Five as a group of homosexuals. Or is it?

GAY might be, as with ENVY a kind of code. If reversed Y - A - G might mean: whY - Anatole Golitsyn whY, or since he and Angleton were good friends: whY - Angleton and Golitsyn.

To understand why this might be, consider the following quote from "Of Moles and Molehunters" found on the CIA's website:

"Important is a definitive chapter in the Schecter-Deriabin book ["The Spy Who Saved the World"], that makes clear that, for the major period of his intelligence production, Penkovsky was not under Soviet control, and his product was not and could not have been deception. The controversy over whether he was bona fide (fueled largely by Angleton and Golitsyn) had arisen after Penkovsky's arrest on 22 October 1962 and was only put to rest within the CIA's Directorate of Operations in 1979 by a long-overdue study of the case. That the Agency made important documentary material available to Schecter and Deriabin so they could provide many of the details to the

public via their excellent book was a laudable action by Dr. Robert Gates, who was DCI at the time."

Also, "The Art of Betrayal" by Gordon Corera, reports that while in London, Golitsyn studied the notes from Penkovsky's case and *"became convinced that Penkovsky was a plant sent as part of a 'master plot' "*, and later: *"After Golitsyn had returned to Washington, he began to pursue the idea that the Soviet Union was undertaking a 'master plan' of 'strategic deception' to fool the West and its intelligence agencies".*

In the wake of the Penkovsky affair it seems that the the CIA and SIS became paranoid about moles within. Golitsyn was able to convince Angleton that more moles existed (Sasha for example). This attitude of conspiracy and paranoia, lead both agencies, and particularly the CIA, to conduct many mole hunts. Many loyal intelligence officers became suspect and were wrongly accused, to the point where some, as is reported in "Of Moles and Molehunters", within the CIA even wondered if Angleton himself was a Soviet Spy. This paranoia hurt the agency, and it wasn't until 1974 when Angleton was removed from the CIA by DCI William Colby that the CIA began to recover.

So the lie that the CIA might have told about Penkovsky was that he was not a genuine agent, but rather a triple agent secretly working for the Soviets as part their "master plan". If they had told Kim Philby this, he and might have (after he defected) threatened to expose it as Philby had seen the truth: that Penkovsky was genuine. Philby could have threatened to expose that the CIA had thought he was a triple too, if the CIA and the SIS exposed any of Bulik's PR. In this case, if Penkovsky was a triple, why would they wish to correct any lie Penkovsky might have told. Apparently, this was the view of the CIA until, after Angleton, conducted a review in 1979 and determined that Penkovsky was genuine. Also, in 1979, Anthony Blunt was revealed to be the "Fourth Man" by Margaret Thatcher.

Whatever the case, Kryptos says Kim sent a message back to the US and Great Britain as to the fate of Penkovsky - an 'X' meaning that Penkovsky had been executed. President JFK and the CIA were distraught about it, but Kryptos seems to say, they had no one to blame but themselves, for it was their plan which had failed.

The actions of the CIA, our shadows, after using the intelligence which Penkovsky gave, brought the world back from the brink of nuclear war. Just a couple of months later, the Soviet's shadow's, the KGB's, "enhanced interrogation techniques" with LSD put into play actions that could have brought on another close brush with nuclear war again. Kryptos, I believe, suggests that the CIA, after finding this out, allowed a lie to dishonor the man who provided that intelligence. Why? Is it for the reason that it considered Penkovsky, for a time, to be a triple agent, or is it because the CIA decided along with the SIS to cover it up and not expose Anthony Blunt, or their own blunder with Kim Philby by sending him with the information from Golitsyn.

Knowing this, Kryptos poses a question - A is X, or A is L; is the Answer the Shadows or is this answer a lie? On the one hand you have the CIA and its agent

Penkovsky, who's actions helped prevented a nuclear war. On the other hand, you had the actions of the KGB which could have lead to the starting of a nuclear war. You have the the mole hunts of the CIA, wrongly accusing Penkovsky and many other loyal officers of being Soviet agents. You have the cover up by the CIA and the SIS of Blunt. Kryptos strongly suggests the former, and that ENVY, Kim Philby's answer, is a lie. It promotes GAY, that Golitsyn and the conspiracies he spread were the reasons why Penkovsky's lie and the actions around it were not exposed. As it was part of the information from these conspiracies: the incomplete information of the five, and the idea that Penkovsky was a triple, which may have lead to Kim Philby to chosen for and to be trusted with this information by Dulles and Angleton. And due to this, it was his, Golitsyn's, ideas about Penkovsky that doomed Penkovsky as well as the mistakes of Angleton and Dulles concerning sending Philby to see if those ideas had any validity.

Again, for A is X, I would like to give a another quote from Joe Bulik, one of Penkovsky's CIA handlers, said in the interview about the information that the spy gave and the CIA during the crisis:

"... Bobby Kennedy said that the Cuban Missile Crisis, which Penkovsky helped us with, was worth all of CIA from 1947 to 1961, all the money that was spent on CIA to that time, it was worth, at least financially that's what it was worth and of course it saved our country. It kept cities from being blown up and it kept the war from going into a nuclear disaster."

I think that these examples, if I am right, show us some important lessons:

That in things relating to nuclear weapons, while our shadows saved us once, too many shadows surrounding them, concerning their numbers, locations and capabilities, (but not how they are built) are a bad idea. Perhaps having them, or at least too many of them, is a bad idea. Jim Sanborn, I believe, hints at this in his work "Cyrillic projector", where the second encrypted document refers to Andrei Sakharov, a Soviet nuclear physicist who in his later years worked towards ending nuclear proliferation.

That, while the business of the shadows will always, as in any war, involve some actions that under normal circumstances are immoral, grossly immoral behavior when committed by them can seriously threaten the very things that they serve to protect, and threaten to undermine their own primary purpose. The CIA was endangered by its own paranoia which led its counterintelligence unit to conduct mole hunts to the point that they resembled witch hunts. Here, it should be noted that William Webster was brought in to help the Agency deal with credibility after the Iran Contra Affair. It seems that Kryptos's subtle message, that witch hunts (mole hunts) like that of the past are harmful to the Agency, might have been meant as a reminder to those seeking to find wrong doing.

The KGB's use of LSD and other immoral interrogation techniques, in this case with Penkovsky, inadvertently could have lead to the start of actions that might have

lead to the destruction of their country. It also shows, at the same time, that the primary functions of our shadows, that of intelligence gathering and interpretation, are critical to the security of our country.

And so perhaps the best realization which can be made about the about Shadows is that the answer is neither true nor lie, black nor white, but the that answer is that Shadows are Grey.

This must be what is buried; documents clearing Penkovsky, hidden gratitude for him from the CIA, and explaining why it was necessary to do what they did. Other documents should be there too; ones that explain their views on Shadows and this event by Kim Philby and George Blake, or maybe Anthony Blunt. Also there might be documents detailing the White House's plan utilizing Kim Philby to save Penkovsky.

However, these document still remain hidden as best as I can tell. I'm not sure that they are still classified since why would Jim know about them, unless he too got security clearance. It seems to me that DCI Webster knew, and helped. I should think that all the directors have known everything about Kryptos except the exact plaintext. To that, they would still have a very good idea. As "Of Moles and Molehunters" says, Bob Gates, the director after Webster, helped release many documents concerning Penkovsky.

I believe that Jim Sanborn and those who helped him are asking us to aid them to clear Penkovsky. To clear the shadows which surround him with those documents. While shadows are undoubtedly necessary, sometimes what is also necessary is that, when they are no longer needed, they are Enlighten for the rest of us to see, for honor to be given or restored, for fault to be better assigned and understood, and for all of us to learn from. As to the call for aid by Jim Sanborn; Jim - I am with you. Whatever I can do to help bring this to light, I will try to do, and I ask all of our Yahoo Group to help Jim in this too. To fix the BUGZ of that great quote on the OHB wall:

**And Ye shall know the truth and
the truth shall set you, Penkovsky, free!**

**Salam and Happy Discoveries and Decrypts,
Ibn Şifre**

References List:

1. "My Silent War; the Soviet Master Spy's Own Story", Kim Philby, 1968
2. "Spy Stories, Of Moles and Molehunters", Cleveland C. Cram, Center for the Study of Intelligence, October 1993, <https://www.cia.gov/library/center-for-the-study-of-intelligence/kent-csi/vol38no5/html/v38i5a15p.htm>
3. "Defend the Realm: The Authorized History of MI5", Christopher Andrew, Random House LLC, 2009 pages 439-440
4. "The Spy Who Saved the World: How a Soviet Colonel Changed the Course of the Cold War", Jerrold L. Schecter and Peter S. Deriabin, Brassey's UK, Limited, 1995
5. "Fifteen DCIs' First 100 Days, Taking stock", CIA History Staff
<https://www.cia.gov/library/center-for-the-study-of-intelligence/csi-publications/csi-studies/studies/95unclass/100Days.html>
6. "The Art of Betrayal", Gordon Corera, Open Road Media, 2013
7. "Dead Drop: The True Story of Oleg Penkovsky and the Cold War's Most Dangerous Operation", Jeremy Duns, Simon and Schuster, 2013
8. "The Spy Who Saved The World—Then Tried To Destroy It", Jeremy Duns, The Daily Beast
<http://www.thedailybeast.com/articles/2013/11/03/the-spy-who-saved-the-world-then-tried-to-destroy-it.html>
9. "The National Security Archive" at The George Washington University, Interview with Joe Bulik, 1998
<http://www2.gwu.edu/~nsarchiv/coldwar/interviews/episode-21/bulik1.html>

Rocks - Spoiler Alert - try to figure it out on your own, maybe I missed something!

Your still reading?! - Oh I know, you want to know where the treasure is buried! I don't know - but I can guess. Are you sure you don't want to try and find out for yourself? If so then stop reading.

Note: Please don't go try and dig anything up at Langley without permission from the DCIA, the Director of the CIA. At the very least, you would be disturbing their Art, and just as the Louvre would not like someone doing anything to the "Mona Lisa", I can't imagine that the CIA would take kindly such actions. Besides, I would guess that the DCIA knows exactly what is buried where and has the unencrypted originals.

From K4 O-PT we can guess that there are 'V L ENCR BURIED': five letters, or documents, encrypted and buried. That would suggest five different locations.

1: Lucid Memory - from Kim Philby

The clues suggest to encrypt our K4 HX-PT with the letter 'X' and look near the end, or back of a reference to Allen Dulles: 'FIND BY ANORM BERLIN HEAD EXP BCK'

K4-HX deciphers to:

D SIS LLL VS O-H-WYE ON BH ZERO ABUZZ X
FAN B CQD, FIND BY ANORM BERLIN HD, EXP BCK - JIM S
T REF, GS MARK IS WD X
ODD RIDDLE IO IS Q J

So if you encrypt HX with a X:

B TGT III QT P-F-UKC PL SF WCZP OSNWW V
DOL S AMB, DGLB SK OLPZJ SCZIGL FB, CVR SAX - HGJ
T Y Z CD, ET JOZX GT UB V
PBB ZGBBIC GP GT M H

and look at the end (back) of the first reference to Allen Dulles, BH for BERLIN HEAD,
BHZER - OABUZZXFANBCQ encrypts to POSNWWVDOLSAM

and that is POS NWW V DO L SA M

If you look at a Position North West by West from T (T is your position and reference) to the other rock, you find V L, a group of five letters, the L and M groups - i.e "LUCID - MEMORY"

Take the AISX key word and encrypt LUCID:

Enc(AISX, LUCID) = **X**DIGL
or X (X marks the spot) DIG (dig here) L (at the L)

Two: The CIA's interpretation

From the P PT:

CIA POSition IN South WorD Vs Lucid Memory

If one goes due south on the compass rose it points to the letter T of

DIGETAL INTERPRETAT**TU**

Note: Digital Interpretation does not mean binary. Digital, from digits, from latin digitus. A 'digit' is a written symbol for any of the numbers 0 to 9. So digital is something relating to ten.

Digital -> of ten -> of Shadows
Digital Interpretation -> Shadow's Interpretation.

One can use the AISX key word and get:
Encode(V, AISX) = PEYQ or better: Q YE P (reversed)

Use the QYEP key:

Enc(QYEP, DIGETAL EEE INTERPRETATU) = TLXHCY HOG KVGFSUGIDXW

Notice the last 5 letters reversed are: W X DIG
with the T decoding to X (again X marks the spot):

However some might, like Kim, might disagree:

Dec(DICUL, LXHCY HOG KVGFSUGIDXW) = ACOPMM ZVM CFCTLIEXSIAG

or at the end:
T LIE XSIA G

Three: documents of George Blake (or KGB)

DIG E T

Note that:

Dec(T, DIGETAL...) = SECAKPG ... SEC A KP, G

And the hint in the O_b PT

BINS -> B IN S -> B IN SW

Which would give us:

T IS YOUR POSITION **E**

Dec(T, T IS YOUR POSITION E) = KEYXRLWZRYEKERIA
KEY X R L, W TRY

From steganography of letter O: P OPEN SAY T, S DIG X READ E WR

Enc(XDIG, T IS YOUR POSITION **E**) = YZUIPOJJPTLGIB**X**

Note the last 7 letters reversed:

X BIG L T J

Four: The Shadow Forces

Use the key 'LSD O' but with the L of the 'LSD O' replaced with an O, and reversed to get 'O DSO' (one could also play with the padding of the ciphertext and change the key to DS00 for D, S, and OO, the Shadow Forces: CIA, SIS and double O's)

Enc(ODSO, **S**HADOW EE FORCES EE) = EXGIDA LJ LIAHJJ LJ

or when the plaintext is reversed, at the end:

J L A DIG **X** E

This X corresponds to the H of SHADOW FORCES

Five: Virtually Invisible

Dec(**TEHB**, EE VIRTUALLY E EEEEEEE INVISIBLE) = AK BAWNAZGSH P
AKWPAK REMTMATSW

Note the last part:

REM T M AT SW; remember T L at SW, so that means use XBIG (and offset by 2 i.e. remove the first two EE's).

Enc(**XBIG**, VIRTUALLY E EEEEE**E** INVISIBLE) =

QWJLNISTK N KXCNKX GREYTWWTTC

Note the last part:

X GREY T WW T C

Shadows are X GREY T WW T C

This might be a message from DCI William Webster about Kryptos.

Now take this plaintext, put back in two letters as padding, and use the key GREY but with the last Y changed to an E for expose:

Enc(**GREE**, NK X GREYTWWTTC) = SR C XHFGIDXIQ

or reversed Q I X DIG F HX C

The X corresponds to the letter B of INVISIBLE

Summing up:

Buried at Letters:

L of **L**UCID MEMORY

T of DIGETAL INTERPRETATU

E of T IS YOUR POSITION **E**

H of **S**HADOW FORCES

B of VIRTUALLY ...E INVISIBLE

E of VIRTUALLY ...**E** INVISIBLE

or if we encrypt the last part of the P (like the O_c PT suggests):

Enc(X, ENVY IN MSG JS) = CLQK GL JTE HT

or at the end **L J TEHT**.

Now the last T in TEHT may be changed to an B and then add an E for expose. So that we get the letters L TEHB E.

Lie Shadows - Enlighten (or Expose)